Hangning Zhang

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Education

Hangzhou Dianzi University — Hangzhou, China
B.Eng. in Automation; ***GPA: 4.51/5.00*** — *Sep 2024 – Jun 2028*
Comprehensive Evaluation: ranked 1st in freshman year — *Sophomore*

Coursework: C Programming, Python, Principles of Automatic Control **Certification:** Huawei HCIA-AI-solution
English: CET-4 540, CET-6 458

Awards & Leadership

Competitions: National First Prize, National College Students Intelligent Car Race; Second Prize, Zhejiang Mathematical Modeling Competition.

Research and Innovation: First author of an ICIBA international conference paper; Principal Investigator, National Undergraduate Innovation Training Program "Ruidong Trajectory".

Campus Honors: Zhejiang Provincial Government Scholarship; First-class University Scholarship three times; Merit Student; Outstanding League Member.

Leadership: League Branch Secretary of Automation Class 6; member of the "Shouzheng Class" Leadership Program; Vice President of the Mathematical Modeling Association.

Skills

Programming: C/C++, Python

Deep Learning and Vision: OpenCV, ResNet, SimAM attention, GNNs, CNN/3D CNN, Transformer baselines

Modeling and Engineering: PID control, multi-agent prompting, DXF/PNG processing, edge inference profiling

Research Experience

Scleral Image Analysis for Carotid Plaque Risk Prediction — Advisor: Prof. Jiuwen Cao

- Investigated non-invasive carotid plaque risk prediction from scleral images and proposed a ResNet-based architecture enhanced with parameter-free SimAM attention.
- Designed dynamic cost-sensitive learning for ambiguous diseased/non-diseased classification, reducing missed diagnoses; validated the modules through ablation studies.

L-RDTNet Radar Target Recognition | *IEEE TII, under review* — Advisor: Assoc. Prof. Haicheng Tu

- Built a lightweight L-RDTNet framework for low-altitude radar target recognition, modeling consecutive RD maps through range-Doppler-time structured representations while preserving Doppler-related motion cues.
- Achieved 95.06% accuracy, 95.04% Macro-F1, and 0.9341 MCC with only 8.37M parameters, reaching 103.41 FPS on a Jetson platform.

Ball Robot Navigation and PPO Training | **Research Internship at Zhejiang University** — Advisor: Prof. You Wang

- Built a multi-parallel-environment physics simulation system in Isaac Gym for model-free navigation control of a ball-shaped robot; completed spherical dynamics modeling, torque control, and point-to-point closed-loop navigation baseline validation, achieving stable simulation of 4 environments $\times$ 27 spheres with precise torque-driven control.
- Designed and implemented a reinforcement learning environment for spherical robot navigation, including the observation space, reward function, success criteria, and independent environment reset logic; completed end-to-end PPO smoke training with a standard MLP policy, validating the training pipeline, inference loop, and environment learnability.

Competitions & Projects

National College Students Intelligent Car Race | *National First Prize* — Vision Lead

- Designed an embedded high-speed autonomous driving vision system with OpenCV-based track recognition, centerline fitting, special-element detection, PID steering, and look-ahead speed control.

OminiBerry Strawberry Disease Diagnosis System — OminiBerry

- Developed a prototype decision-support system for strawberry leaf, flower, and fruit disease diagnosis, supporting image upload, visual summarization, knowledge retrieval, multi-expert reasoning, and Markdown report generation.
- Designed agent roles for pathology reasoning, differential exclusion, treatment recommendation, and cultivation management, producing structured and explainable diagnostic outputs through prompt templates and report packaging.
- Built FastAPI backend routes and pipeline modules with case-library, knowledge-base, and run-record persistence; configured lightweight local execution with Ollama to reduce deployment and debugging cost.